

Appendix from ASTRODEEP-JWST: NIRCam-HST multiband photometry and redshifts for half a million sources in six extragalactic deep fields

E. Merlin[1], P. Santini[1], D. Paris[1], M. Castellano[1], A. Fontana[1], T. Treu[2], S. L. Finkelstein[3], J. S. Dunlop[4], P. Arrabal Haro[5], M. Bagley[3], K. Boyett[6,7], A. Calabrò[1], M. Correnti[1,8], K. Davis[9], M. Dickinson[5], C. T. Donnan[4], H. C. Ferguson[10], F. Fortuni[1], M. Giavalisco[11], K. Glazebrook[12], A. Grazian[13], N. A. Grogin[10], N. Hathi[10], M. Hirschmann[14], J. S. Kartaltepe[15], L. J. Kewley[16], A. Kirkpatrick[17], D. D. Kocevski[18], A. M. Koekemoer[10], G. Leung[3], J. M. Lotz[10], R. A. Lucas[10], D. K. Magee[19], D. Marchesini[20], S. Mascia[1], D. J. McLeod[4], R. J. McLure[4], T. Nanayakkara[12], L. Napolitano[1,21], M. Nonino[22], C. Papovich[23,24], L. Pentericci[1], P. G. Pérez-González[25], N. Pirzkal[26], S. Ravindranath[27,28], G. Roberts-Borsani[29], R. S. Somerville[30], M. Trenti[6,7], J. R. Trump[9], B. Vulcani[13], X. Wang[31,32,33], P. J. Watson[13], S. M. Wilkins[34,35], G. Yang[36,37], and L. Y. A. Yung[10]

(Affiliations can be found after the references)

October 4, 2024

ABSTRACT

This document provides the Appendix Section from the paper "ASTRODEEP-JWST: NIRCam-HST multiband photometry and redshifts for half a million sources in six extragalactic deep fields" by E. Merlin et al., 2024, DOI 10.1051/0004-6361/202451409.
The catalogues are available for download from the Astrodeep website `https://astrodeep.eu/astrodeep-jwst-catalogs/`. Updates and/or additional releases will be documented on the website.
It is also possible to visualize and explore the catalogues on the Cosmological Surveys Rainbow Database, `http://arcoirix.cab.inta-csic.es/Rainbow_navigator_public/` (Pérez-González et al. 2005, 2008; Barro et al. 2019).
The "optimal" photometric catalogues (see Sect. 3.4 of the main paper) and the photo-z catalogues (see Section 5) for all fields are also available in electronic form at the CDS via anonymous ftp to cdsarc.u-strasbg.fr (130.79.128.5) or via http://cdsweb.u-strasbg.fr/cgi-bin/qcat?J/A+A/.

Appendix A: Catalogue format

We release several catalogues for each field, among which one (named "optap"-catalogue) is obtained assigning to each source the colours computed in its "optimal" aperture (see Section 3.4). The catalogues are published in `fits` format, and come with a `README` file that explains the meaning and format of the columns. All catalogues first list a set of columns indicating the unique identifiers of the detected objects, their position in equatorial coordinates and in pixels, and some values from the SExtractor detection run on the F356W+F444W stack mosaics: `ISOAREA_IMAGE`, `CLASS_STAR`, `FLAGS`, `FLUX_RADIUS`, `FLUX_AUTO`, `FLUXERR_AUTO`. Then, some measures obtained with a-phot again on the detection stack are listed: the major semi-axis of the elliptical isophote, the ellipticity and the position angle, and the Kron radius (in pixels). The total fluxes in the 16 photometric bands, computed as described in Section 3.5, are then given, followed by the corresponding 16 uncertainties (all in μJy units). Finally, the last column lists the flags discussed in Section 3.5. In the "optap"-catalogue, the value of the "optimal" aperture is also given (in arcseconds). We also release the original raw catalogue containing all the measured aperture fluxes.

We separately release a set of photometric redshift catalogues estimated from the "optap"-catalogue with local background subtraction. For each source we list the ID, RA and Dec, the spectroscopic redshift when available, the four estimates of the photometric redshift (see Section 5), and the flag assigned to the source as reported also in the "optap"-catalogue, with the additional indication of a negative sign for the sources lacking information in the bands necessary to identify the Lyman break.

Appendix B: Astrometry

As explained in Section 3.1, we re-aligned HST mosaics to the NIRCam grid by means of an automatic procedure. Table B.1 lists the offsets in RA and Dec after the procedure, for all the bands and fields. We point out that the CANDELS EGS and CEERS HDR1 HST data are obtained from the same observations, but the latter has been re-aligned to Gaia-DR3; since we took care to correct all of the images for astrometric accuracy, the two datasets should be perfectly consistent.

Table B.1: The median and MAD of the astrometric offsets of the HST bands.

Band	$N_{obj.}$	ΔRA (")	ΔDEC (")
	ABELL2744		
F435W	1307	0.009 ± 0.02	0.000 ± 0.01
F606W	3309	0.001 ± 0.02	0.001 ± 0.01
F814W	3188	-0.001 ± 0.02	-0.001 ± 0.01
F105W	4595	0.003 ± 0.01	0.001 ± 0.01
F125W	2170	0.003 ± 0.02	-0.001 ± 0.01
F140W	849	0.005 ± 0.02	0.003 ± 0.01
F160W	2054	0.003 ± 0.02	0.000 ± 0.01
	CEERS		
F606W	7063	0.044 ± 0.03	0.032 ± 0.02
F814W	8615	0.066 ± 0.02	0.041 ± 0.01
F105W	1437	0.007 ± 0.05	0.010 ± 0.03
F125W	8470	0.027 ± 0.01	0.036 ± 0.01
F160W	4355	0.064 ± 0.05	0.040 ± 0.02
	JADES-GN		
F435W	1144	0.036 ± 0.05	0.015 ± 0.02
F606W	1822	0.045 ± 0.04	0.015 ± 0.02
F814W	3228	0.040 ± 0.03	0.015 ± 0.01
F105W	2800	0.041 ± 0.02	0.012 ± 0.01
F125W	3430	0.039 ± 0.01	0.013 ± 0.00
F140W	1532	0.044 ± 0.01	0.015 ± 0.01
F160W	1436	0.041 ± 0.04	0.013 ± 0.02
	JADES-GS		
F435W	13997	0.006 ± 0.03	-0.015 ± 0.03
F606W	21408	0.004 ± 0.02	-0.012 ± 0.02
F814W	19677	0.000 ± 0.02	-0.010 ± 0.01
F105W	12169	0.005 ± 0.01	-0.013 ± 0.01
F125W	18764	0.007 ± 0.01	-0.013 ± 0.01
F140W	8542	0.008 ± 0.01	-0.014 ± 0.01
F160W	6235	0.006 ± 0.03	-0.013 ± 0.02
	NGDEEP		
F435W	758	0.082 ± 0.04	-0.058 ± 0.03
F606W	1194	0.077 ± 0.03	-0.052 ± 0.03
F775W	1121	0.077 ± 0.03	-0.052 ± 0.03
F814W	1096	0.079 ± 0.03	-0.053 ± 0.02
F105W	815	0.091 ± 0.01	-0.049 ± 0.01
F125W	965	0.091 ± 0.01	-0.050 ± 0.01
F160W	506	0.089 ± 0.02	-0.049 ± 0.02
	PRIMER-COSMOS		
F435W	733	0.010 ± 0.04	0.012 ± 0.05
F606W	1279	-0.011 ± 0.03	-0.022 ± 0.03
F814W	2925	-0.004 ± 0.01	-0.008 ± 0.01
F125W	4704	-0.004 ± 0.01	-0.011 ± 0.01
F140W	2797	-0.006 ± 0.01	-0.006 ± 0.02
F160W	3709	-0.005 ± 0.02	-0.011 ± 0.02
	PRIMER-UDS		
F435W	1214	0.008 ± 0.04	0.000 ± 0.04
F606W	3613	-0.003 ± 0.03	0.010 ± 0.03
F814W	4779	-0.003 ± 0.03	0.009 ± 0.03
F125W	3911	0.010 ± 0.01	-0.004 ± 0.01
F140W	1653	0.010 ± 0.02	-0.004 ± 0.02
F160W	4915	0.010 ± 0.03	-0.005 ± 0.04

Notes. The values are computed by cross-matching N_{obj} sources and corrected before re-projecting the HST images to the JWST grid. For each field, the F160W values are the coordinate offsets between the original HST F160W and the NIRCam F150W band, while the others refer to the offsets between the given band and F160W band after it was re-aligned to the F150W.

Appendix C: Selection of point-like and potentially spurious sources

The PCA technique described in Section 3.5 works as follows. We fed the `scikit-learn` module `decomposition.PCA` with three parameters from the SExtractor detection catalogue, i.e. the signal-to-noise ratio S/N obtained as the ratio between `FLUX_AUTO` and `FLUXERR_AUTO`, the peak surface brightness above background `MUMAX`, and the half light radius `FLUX_RADIUS`. We used a sub-sample of the PRIMER-COSMOS SExtractor detection catalogue to determine the principal components matrix [[0.38283975 0.65126842 0.65519704], [0.92363354 -0.28388812 -0.25750461]] (with explained variance ratio [0.65961785 0.27738227]), and then applied this matrix to the full catalogues of all fields. The loci of PLS and PSS are easily identified in the resulting principal components diagram, and after a rigid anti-clockwise rotation of 66.5° PSS can be singled out by a simple law, $0.15 < \mathtt{rPCA2} < 0$ & $\mathtt{rPCA2} < c_{st}$ (where `rPCAi` indicates the value of the rotated component). PSS can be identified as those in the uppermost region of the diagram, using the law $\mathtt{rPC2} > c_{sp1} \times \log_{10}(1 + \mathtt{rPC1}) + c_{sp2}$ & $\mathtt{rPC1} > c_{sp3}$. The exact definition of the locus of the two populations slightly varies from field to field, as shown by the values of the constants reported in Table C.1.

We point out that this approach joins two similar and complementary techniques using diagnostic diagrams to identify the loci occupied by point-like and spurious sources, i.e. the S/N vs radius plane and the μ-mag vs mag plane. Fig. C.1 shows the PCA plane for all fields in the left column, and the two diagrams corresponding to the other techniques in the central and right columns.

Table C.1: Constants used to single out stars and potentially spurious objects in the rotated PCA plane.

Field	c_{st}	c_{sp1}	c_{sp2}	c_{sp3}
ABELL2744	0.5	0	1.75	-0.2
CEERS	0.5	3	2	-10
JADES-GN	0.8	3	2	-10
JADES-GS	0.5	5	2.5	-10
NGDEEP	0.8	3	2	-10
PRIMER-COSMOS	0.5	0	2.2	-0.2
PRIMER-UDS	0.5	0	2	-10

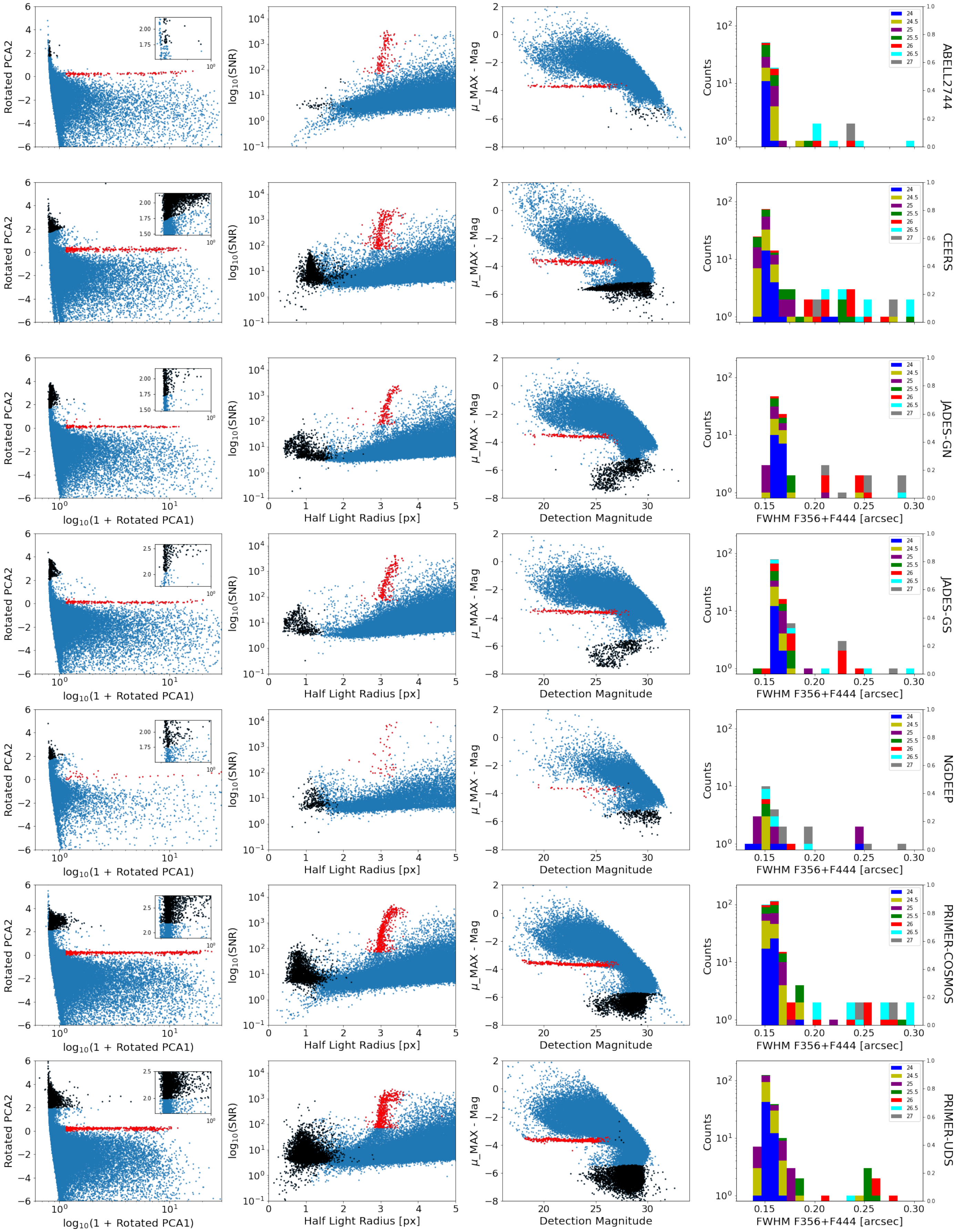

Fig. C.1: Star/galaxy separation and flagging of spurious sources. From left to right: as obtained in the PCA plane described in Section 3.5, projected on the hlr vs. S/N plane, projected on the μMAX-MAG vs. MAG plane. The last column shows the values of the FWHM in the detection F356W+F444W stack, for the sources identified as point-source.

Appendix D: Comparisons with other photometric catalogues

Fig. D.1 to D.6 show how the photometry obtained in this work compares to that from other available catalogues. For each field we consider recent NIRCam-based catalogues and archival HST based data, from CANDELS or Astrodeep. For the latter, we used the Ks and IRAC CH1-2 bands as proxies for the F200W, F356W and F444W JWST bands, applying the following colour corrections, based on Bruzual & Charlot (2003) theoretical templates: Ks=F200W-0.06, IRAC1=F356W+0.02, IRAC2=F444W-0.01.

Each panel of the plots shows the comparison for one colour, with the relative difference $\Delta c/c$ (i.e. colour measured in this work minus colour in the reference work, divided by colour in the archival work) as a function of the magnitude of the second band in this work (e.g., F444W magnitude if the colour is F356W-F444W). The blue line is the median of the distribution. We consider sources with S/N>5 and flag<200. See Section 4 for more details.

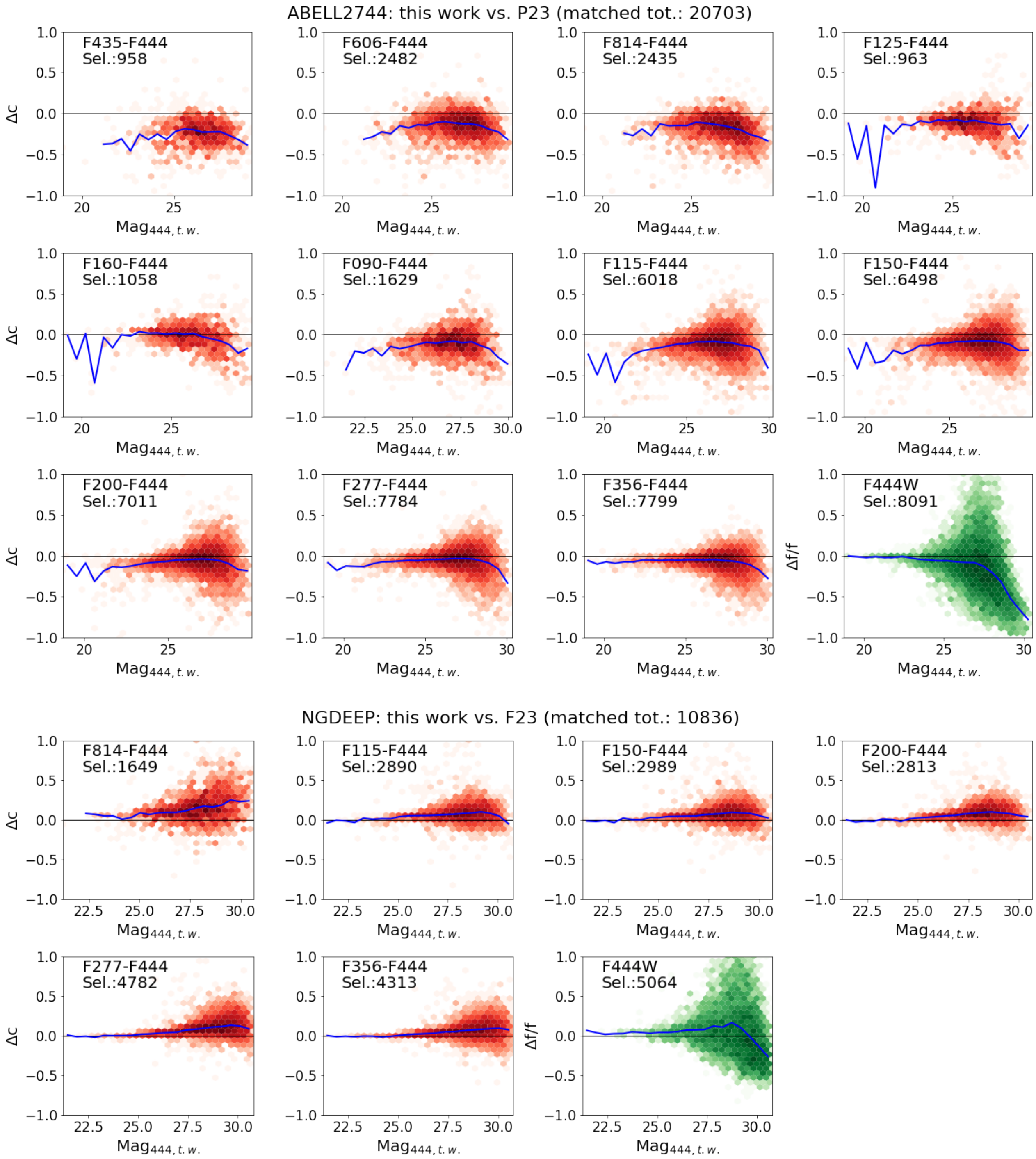

Fig. D.1: Comparison of colours measured in this work and archival catalogues. Shown are relative errors Δc, i.e. the colours measured in this work minus those in the reference catalogues), vs. the second band magnitude in this work catalogue (e.g. F444W for the F356-F444W colour). Top to bottom: ABELL2744 vs. P23 and NGDEEP vs. Finkelstein's catalogue (priv. comm.). The number of cross-matched sources after excluding those with S/N < 5 in any of the two catalogues or flag ≥ 200 in this work is also given in each panel; the blue line is the median of the distribution.

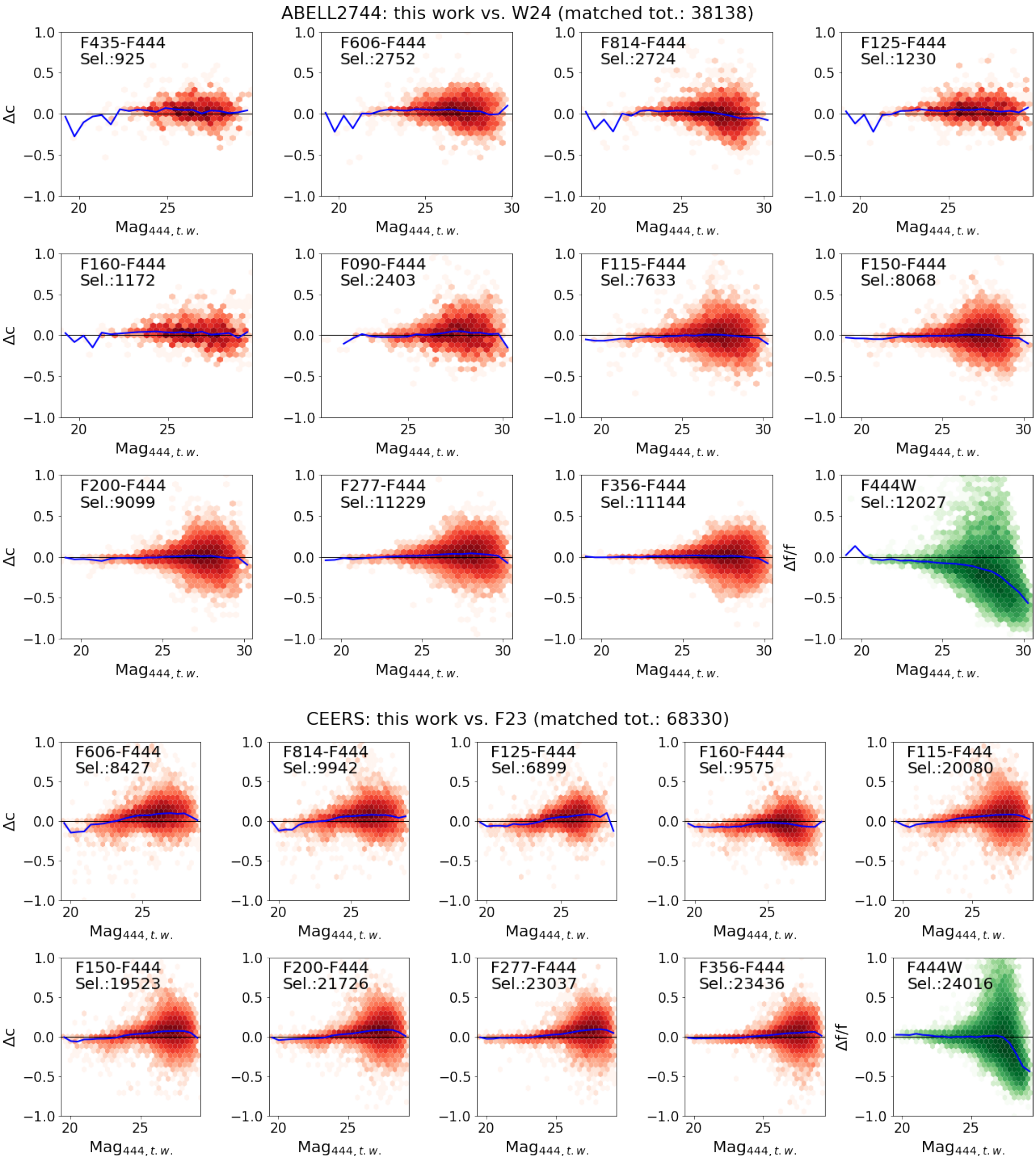

Fig. D.2: Same as Fig. D.1, for (top to bottom) ABELL2744 vs. Weaver et al. (2024) and CEERS vs. Finkelstein et al. (2023).

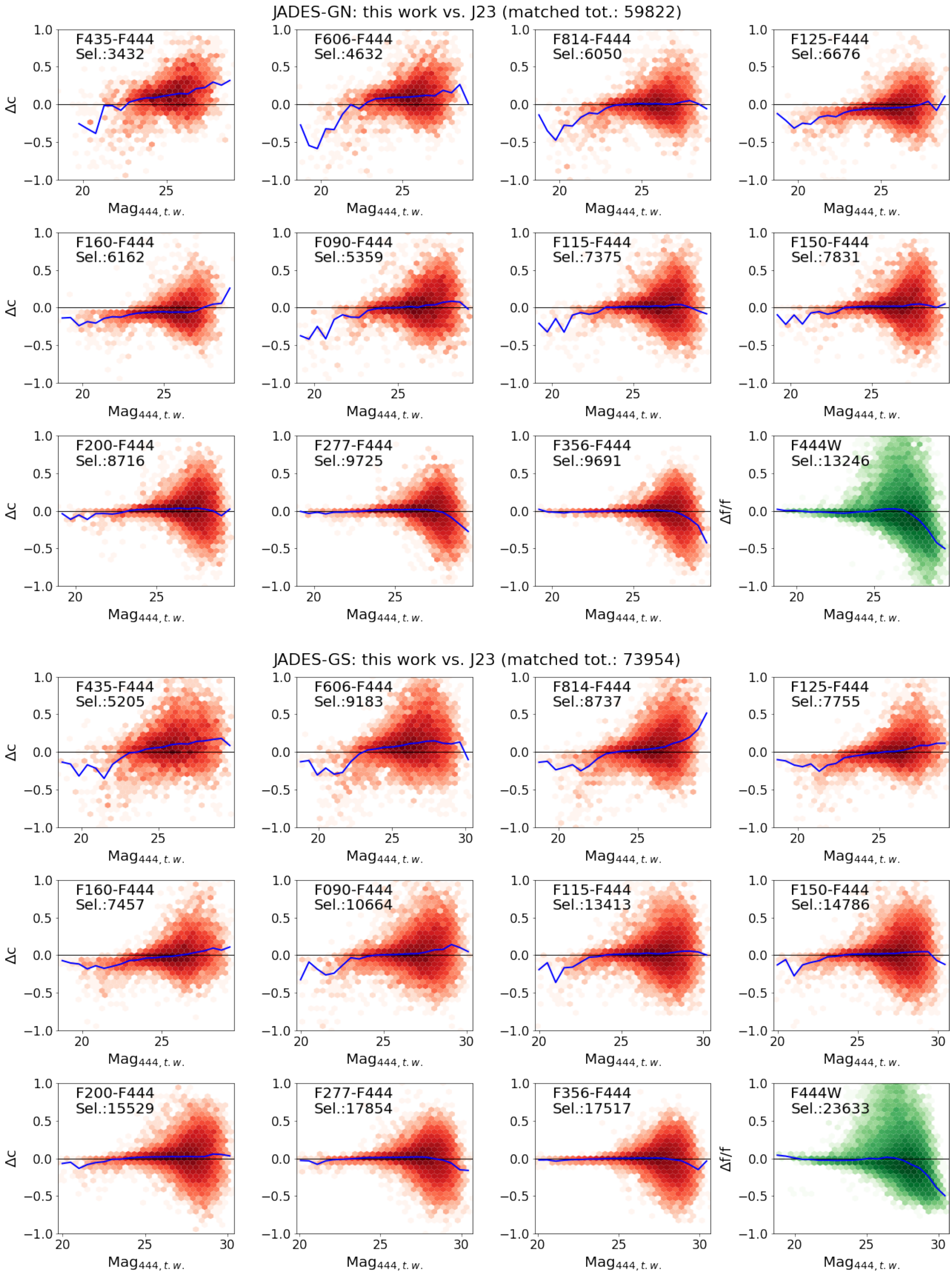

Fig. D.3: Same as Fig. D.1, for JADES-GN and JADES-GS vs. the JADES team catalogues (Rieke et al. 2023).

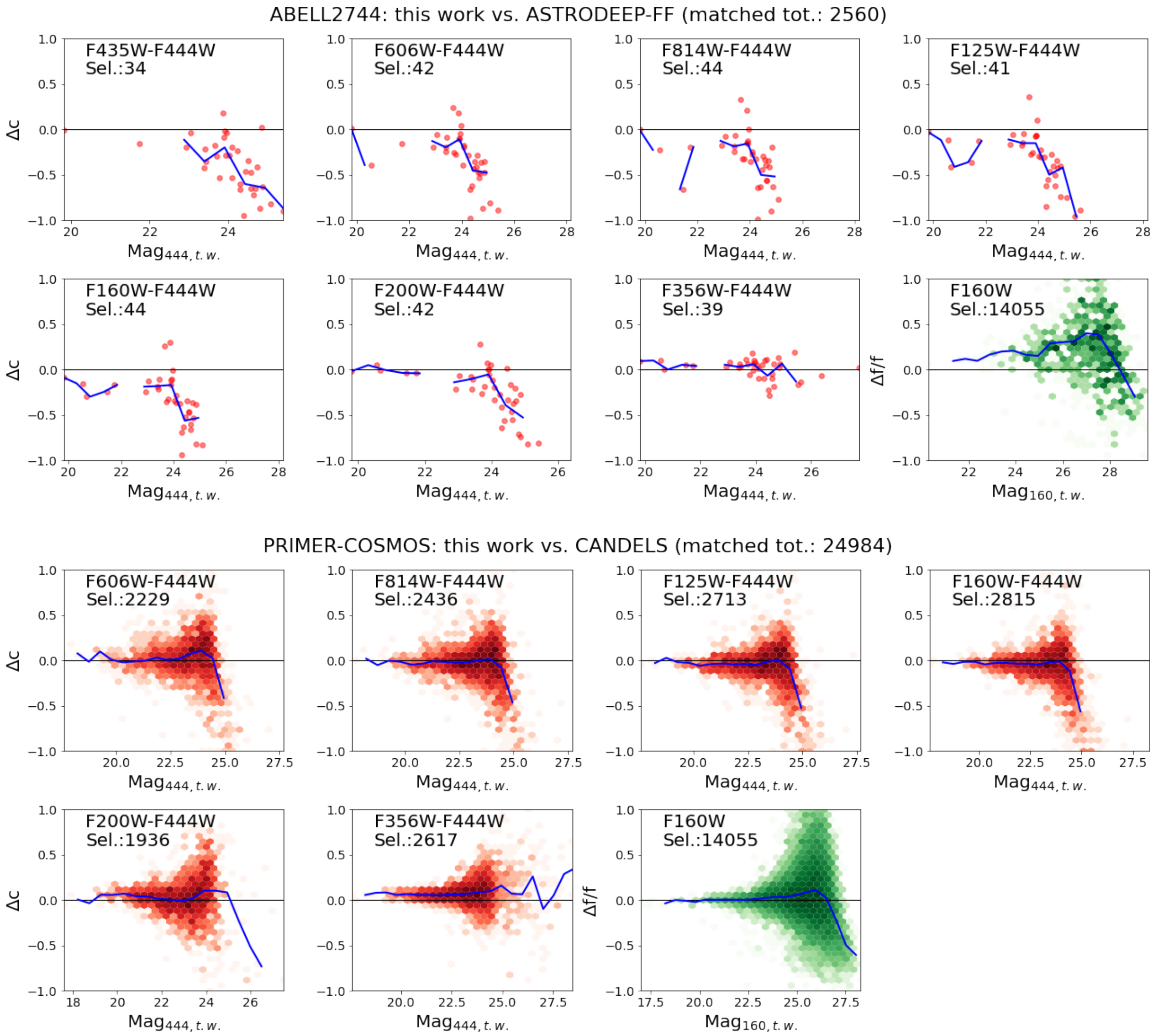

Fig. D.4: Same as Fig. D.1, for ABELL2744 vs. Merlin et al. (2016) showing individual points rather than a density `hexbin` plot because of the small number of cross-matched sources, and PRIMER-COSMOS vs. Nayyeri et al. (2017). The HST colours are transformed into JWST colours by means of the corrections described in Appendix D.

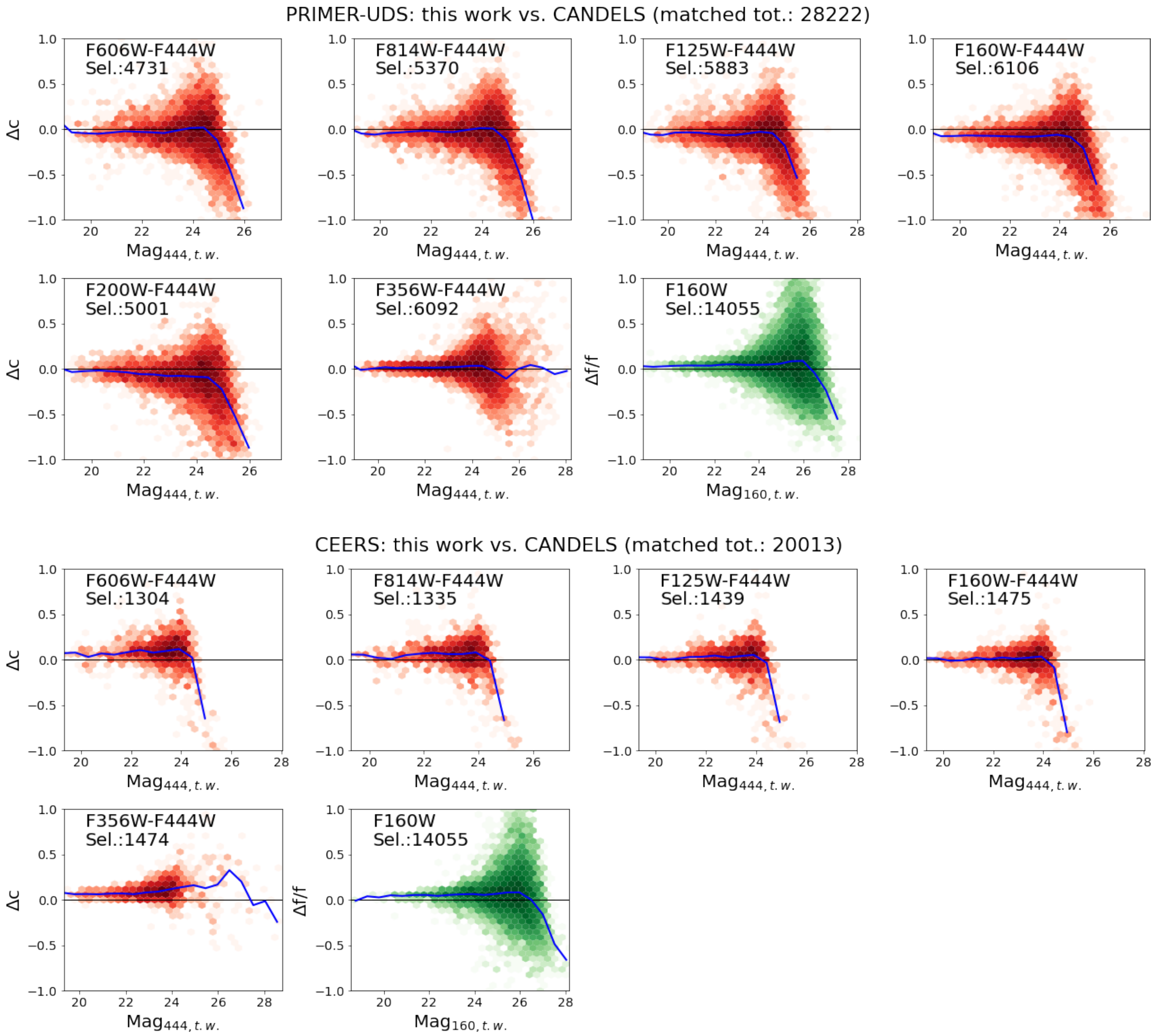

Fig. D.5: Same as Fig. D.1, for PRIMER-UDS vs. Galametz et al. (2013), and CEERS vs. Stefanon et al. (2017).

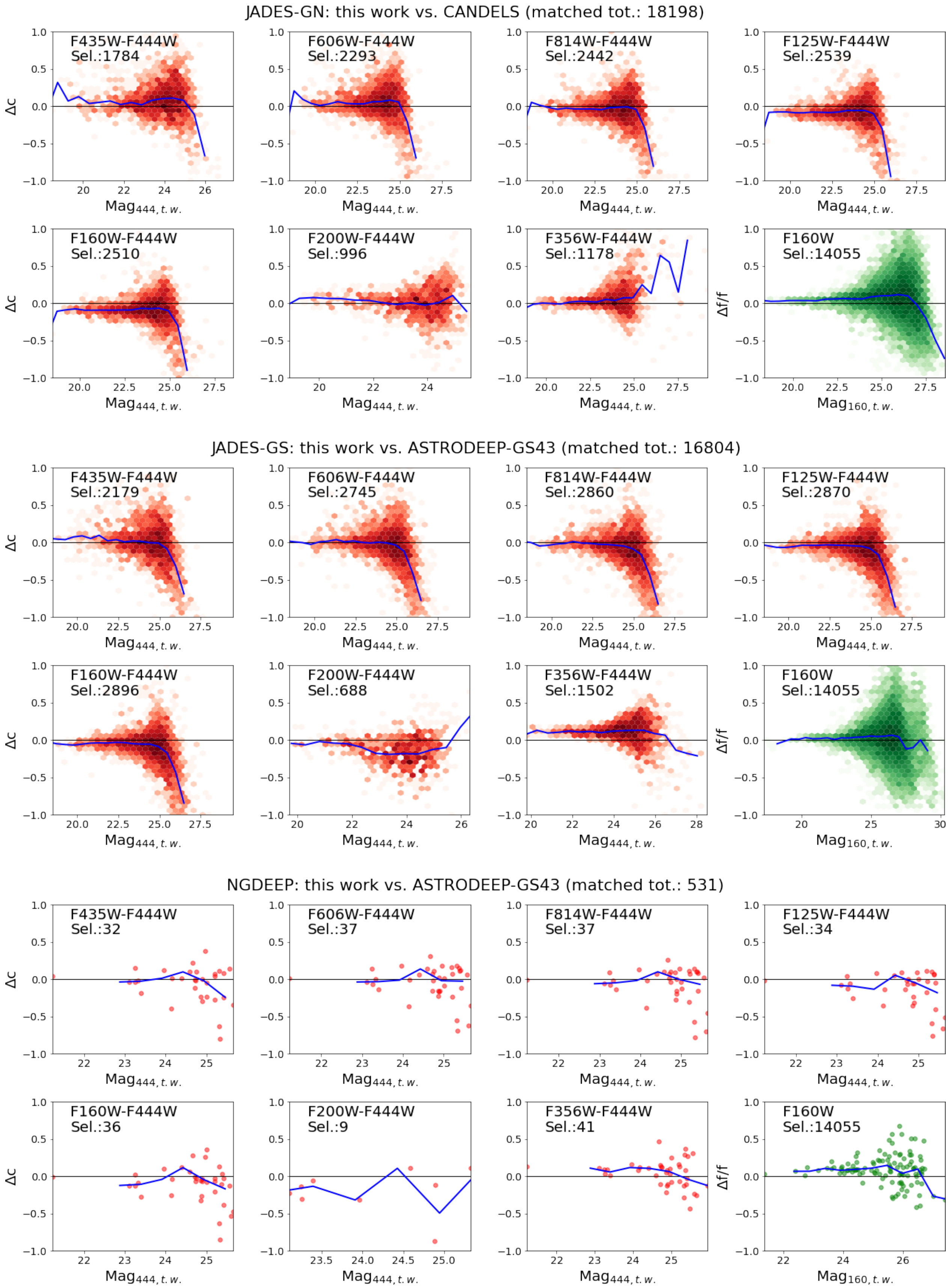

Fig. D.6: Same as Fig. D.1, JADES-GS vs. Merlin et al. (2021) and NGDEEP vs. Merlin et al. (2021), the latter showing individual points rather than a density `hexbin` plot because of the small number of cross-matched sources. The HST colours are transformed into JWST colours by means of the corrections described in Appendix D.

Appendix E: Photometric redshift validation

In Fig. E.1 we show the comparison between the spectroscopic and photometric redshifts, the latter computed as the median of the three runs with (i) ZPHOT, (ii) EAzY with `eazy_v1.3`, and (iii) EAzY with Larson et al. (2023) FSPS Set 1 + Set 4 templates, as described in Section 5. Table E.1 summarises the statistics for all four runs.

Table E.1: Comparative statistics for photometric vs. spectroscopic redshifts.

	N	100×mean	100×stdev	100×median	100×NMAD	η
			ABELL2744			
ZPHOT	1355	0.57	4.28	0.47	3.33	10.04
EAzY v1.3	1354	-0.81	4.31	-0.46	3.36	13.81
EAzY Larson Lya	1338	-0.15	4.64	-0.28	3.29	13.75
EAzY Larson LyaRed	1338	-0.32	4.70	-0.33	3.46	14.28
			CEERS			
ZPHOT	3186	1.17	4.63	0.89	3.64	7.97
EAzY v1.3	3182	0.38	4.55	0.41	3.59	7.54
EAzY Larson Lya	3180	-0.40	4.96	-0.26	4.04	9.03
EAzY Larson LyaRed	3180	-0.41	4.97	-0.30	4.08	9.18
			JADES-GN			
ZPHOT	1848	0.81	4.25	0.75	3.47	6.71
EAzY v1.3	1846	-0.24	4.30	0.01	2.98	8.88
EAzY Larson Lya	1841	-0.74	4.08	-0.53	2.89	8.53
EAzY Larson LyaRed	1841	-0.77	4.10	-0.55	2.93	8.75
			JADES-GS			
ZPHOT	2021	0.41	4.01	0.32	3.02	5.79
EAzY v1.3	2019	-0.57	4.05	-0.35	2.91	10.10
EAzY Larson Lya	2017	-0.75	3.80	-0.47	2.81	9.92
EAzY Larson LyaRed	2017	-0.85	3.83	-0.54	2.84	10.21
			NGDEEP			
ZPHOT	128	1.46	4.08	0.92	3.10	10.16
EAzY v1.3	128	-0.10	4.26	0.21	3.62	9.38
EAzY Larson Lya	128	-0.29	4.16	-0.14	3.76	15.62
EAzY Larson LyaRed	128	-0.29	4.18	-0.10	3.80	16.41
			PRIMER-COSMOS			
ZPHOT	4413	1.37	4.61	1.32	4.15	6.80
EAzY v1.3	4408	-0.03	4.59	0.10	3.64	7.17
EAzY Larson Lya	4399	-0.63	4.69	-0.36	3.85	8.98
EAzY Larson LyaRed	4399	-0.65	4.71	-0.37	3.89	9.09
			PRIMER-UDS			
ZPHOT	3721	1.03	4.50	0.95	3.80	5.80
EAzY v1.3	3719	-0.05	4.55	0.09	3.65	5.22
EAzY Larson Lya	3713	-0.64	4.86	-0.43	3.85	7.41
EAzY Larson LyaRed	3713	-0.65	4.86	-0.46	3.87	7.57

Notes. The listed values are: number N of matched sources (the number can vary due to failings in the fitting procedures of the different codes); mean, standard deviation, median and NMAD of the quantity $|z_{phot} - z_{spec}|/(1 + z_{spec})$, all multiplied by 100 to make the differences clearer; and percentage of outliers $f_{outliers}$ as described in Section 5, for each run using different software and/or templates, and for each field.

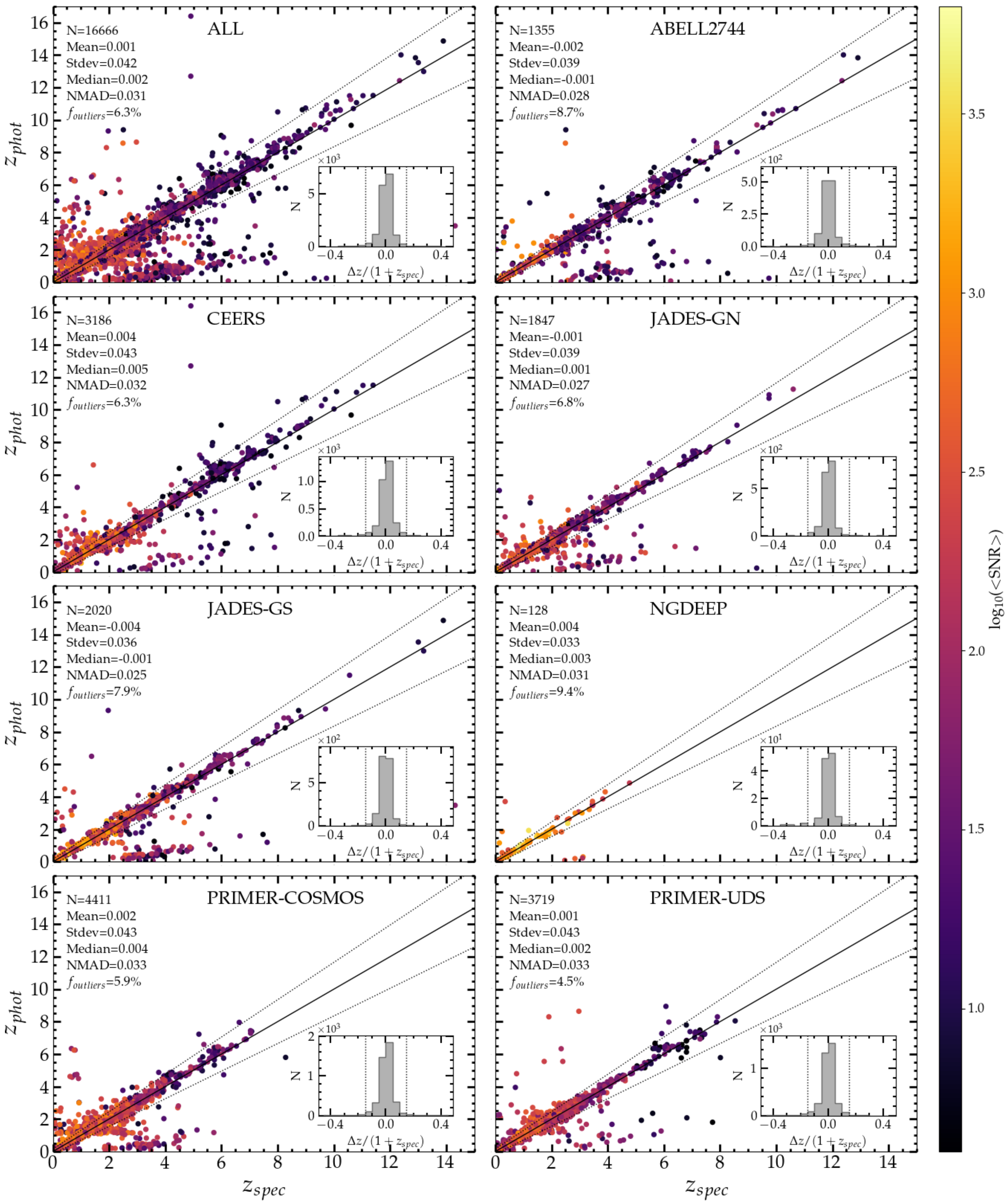

Fig. E.1: Accuracy of photometric redshifts (median of the three runs described in Section 5) on spectroscopic samples, for all fields together (top left panel), and separately for each field. In each panel, N is the number of sources; $f_{outliers}$ is the percentage of sources with $|z_{phot} - z_{spec}|/(1 + z_{spec}) > 0.15$; the other statistics are computed on non-outliers, with NMAD=$1.48 \times \mathrm{median}[|z_{phot} - z_{spec}|/(1 + z_{spec})]$. colour coding is $< S/N > = (S/N_{F444W} + S/N_{F356W} + S/N_{F277W} + S/N_{F200W})/4$.

References

Barro, G., Pérez-González, P. G., Cava, A., et al. 2019, ApJS, 243, 22
Bruzual, G. & Charlot, S. 2003, MNRAS, 344, 1000
Finkelstein, S. L., Bagley, M. B., Ferguson, H. C., et al. 2023, ApJ, 946, L13
Galametz, A., Grazian, A., Fontana, A., et al. 2013, ApJS, 206, 10
Larson, R. L., Hutchison, T. A., Bagley, M., et al. 2023, ApJ, 958, 141
Merlin, E., Bourne, N., Castellano, M., et al. 2016, A&A, 595, A97
Merlin, E., Castellano, M., Santini, P., et al. 2021, A&A, 649, A22
Nayyeri, H., Hemmati, S., Mobasher, B., et al. 2017, ApJS, 228, 7
Pérez-González, P. G., Rieke, G. H., Egami, E., et al. 2005, ApJ, 630, 82
Pérez-González, P. G., Rieke, G. H., Villar, V., et al. 2008, 675, 234
Rieke, M. J., Robertson, B., Tacchella, S., et al. 2023, ApJS, 269, 16
Stefanon, M., Yan, H., Mobasher, B., et al. 2017, ApJS, 229, 32
Weaver, J. R., Cutler, S. E., Pan, R., et al. 2024, ApJS, 270, 7

---

[1] INAF – Osservatorio Astronomico di Roma, Via Frascati 33, 00078 Monteporzio Catone, Rome, Italy
e-mail: emiliano.merlin@inaf.it
[2] Department of Physics and Astronomy, University of California, Los Angeles, 430 Portola Plaza, Los Angeles, CA 90095, USA
[3] Department of Astronomy, The University of Texas at Austin, Austin, TX, USA
[4] SUPA[1], Institute for Astronomy, University of Edinburgh, Royal Observatory, Edinburgh EH9 3HJ, UK
[5] NSF's National Optical-Infrared Astronomy Research Laboratory, 950 N. Cherry Ave., Tucson, AZ 85719, USA
[6] School of Physics, University of Melbourne, Parkville 3010, VIC, Australia
[7] ARC Centre of Excellence for All Sky Astrophysics in 3 Dimensions (ASTRO 3D), Australia
[8] Space Science Data Center, Italian Space Agency, via del Politecnico, 00133, Roma, Italy
[9] Department of Physics, 196A Auditorium Road, Unit 3046, University of Connecticut, Storrs, CT 06269, USA[2]
[10] Space Telescope Science Institute, Baltimore, MD, USA
[11] University of Massachusetts, 710 N. Pleasant St, LGRB-520, Amherst, MA 01003, USA
[12] Centre for Astrophysics and Supercomputing, Swinburne University of Technology, PO Box 218, Hawthorn, VIC 3122, Australia
[13] INAF – Osservatorio Astronomico di Padova, Vicolo Osservatorio 5, 35122 Padova, Italy
[14] Institute of Physics, Lab for Galaxy Evolution, EPFL, Observatoire de Sauverny, Chemin Pegasi 51, 1290 Versoix, Switzerland
[15] Laboratory for Multiwavelength Astrophysics, School of Physics and Astronomy, Rochester Institute of Technology, 84 Lomb Memorial Drive, Rochester, NY 14623, USA
[16] Center for Astrophysics | Harvard & Smithsonian, 60 Garden Street, Cambridge, MA 02138, USA
[17] Department of Physics and Astronomy, University of Kansas, Lawrence, KS 66045, USA
[18] Department of Physics and Astronomy, Colby College, Waterville, ME 04901, USA
[19] Department of Astronomy and Astrophysics, UCO/Lick Observatory, University of California, Santa Cruz CA 95064, USA
[20] Department of Physics & Astronomy, Tufts University, MA 02155, USA
[21] Dipartimento di Fisica, Università di Roma Sapienza, Città Universitaria di Roma - Sapienza, Piazzale Aldo Moro, 2, 00185, Roma, Italy
[22] INAF – Osservatorio Astronomico di Trieste, Via Tiepolo 11, I-34131 Trieste, Italy
[23] Department of Physics and Astronomy, Texas A&M University, College Station, TX, 77843-4242 USA
[24] George P. and Cynthia Woods Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX, 77843-4242 USA
[25] Centro de Astrobiología (CAB), CSIC-INTA, Ctra. de Ajalvir km 4, Torrejón de Ardoz, E-28850, Madrid, Spain
[26] ESA/AURA Space Telescope Science Institute
[27] Astrophysics Science Division, NASA Goddard Space Flight Center, 8800 Greenbelt Road, Greenbelt, MD 20771, USA
[28] Center for Research and Exploration in Space Science and Technology II, Department of Physics, Catholic University of America, 620 Michigan Ave N.E., Washington DC 20064, USA
[29] Department of Astronomy, University of Geneva, Chemin Pegasi 51, 1290 Versoix, Switzerland
[30] Center for Computational Astrophysics, Flatiron Institute, 162 5th Avenue, New York, NY 10010, USA
[31] School of Astronomy and Space Science, University of Chinese Academy of Sciences (UCAS), Beijing 100049, China
[32] National Astronomical Observatories, Chinese Academy of Sciences, Beijing 100101, China
[33] Institute for Frontiers in Astronomy and Astrophysics, Beijing Normal University, Beijing 102206, China
[34] Astronomy Centre, University of Sussex, Falmer, Brighton BN1 9QH, UK
[35] Institute of Space Sciences and Astronomy, University of Malta, Msida MSD 2080, Malta
[36] Kapteyn Astronomical Institute, University of Groningen, P.O. Box 800, 9700 AV Groningen, The Netherlands
[37] SRON Netherlands Institute for Space Research, Postbus 800, 9700 AV Groningen, The Netherlands